Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A Translatable Skin Recovery Platform Using Anti-Inflammatory Natural Extracts Delivered via Patch			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio and Medical		Bio and Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Drug Delivery Systems	Industrial Biotechnology	Functional Biomaterials
Concept of Proposed Technology	• The proposed technology aims to establish a scalable skin recovery platform that integrates scientifically supported anti-inflammatory natural actives into optimized patch- and microneedle-based delivery systems. The platform is designed to provide a practical on-skin solution that can rapidly reduce irritation-associated skin responses while supporting recovery of stressed or mildly damaged skin. Importantly, the proposed concept is not intended to produce a single fixed formulation, but rather to build a modular development platform that can be adapted across multiple product formats and user scenarios while maintaining consistent scientific and engineering rationale. • Skin irritation and inflammation-related responses are not limited to cosmetic discomfort. They arise broadly from daily-life exposure and repetitive stress conditions including friction, dryness, minor abrasion, thermal stress, environmental pollutants, and procedure-related irritation. These factors often lead to redness, discomfort, sensitivity, and delayed barrier recovery, which can negatively affect skin function and quality of life (1). While topical products such as creams or gels are widely used for soothing and recovery purposes, their real-world performance is frequently limited by short residence time on skin, inconsistent dosing, and variability in user application behavior. In addition, natural extracts often present formulation challenges such as stability loss during storage, undesired phase changes in polymeric matrices, and batch-to-batch variability, which together			

reduce product reliability and hinder scalable translation into market-ready prototypes (2, 3).

- To address these limitations, the central concept of this project is to engineer a modular natural-active delivery platform that integrates formulation design, device architecture, and standardized performance evaluation in a single framework. From an engineering perspective, the proposed technology will combine microfabricated adhesive flat patches and dissolving microneedle patch formats as complementary delivery architectures. Film- or hydrogel-based patches will be used to provide localized soothing and barrier-support functionality through optimized polymer matrices and a controlled skin-contact interface. Such patch formats can create a protective microenvironment that supports recovery by maintaining hydration, reducing exposure to secondary irritation, and enabling prolonged local interaction of active ingredients with the skin surface. In parallel, dissolving microneedle patches will serve as an advanced option designed to improve delivery efficiency by enabling controlled interaction beyond the stratum corneum, while remaining minimally invasive and compatible with practical, user-friendly application (4,5).

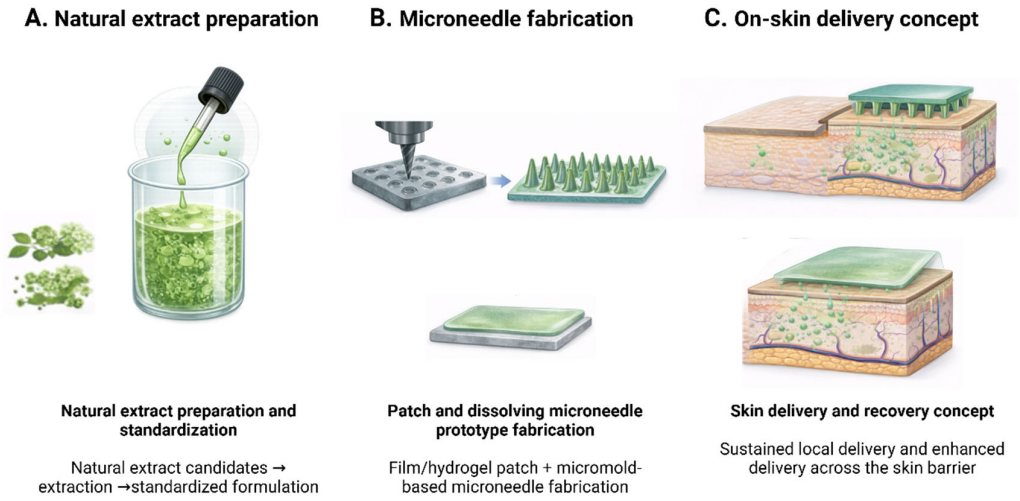

Fig 1. Concept schematic of the proposed skin recovery platform integrating anti-inflammatory natural extract into patch- and microneedle-based delivery formats.

- The proposed platform will emphasize rational material selection and formulation engineering to ensure that prototypes are not only biologically effective but also technically feasible as real products. Key design considerations will include stability of natural actives within polymer matrices, prevention of undesired crystallization or phase separation, and incorporation of protective or passivation strategies where required to preserve marker integrity during processing and storage. Patch architecture will also be optimized to support controlled release behavior and consistent delivery performance, while ensuring wearability-related characteristics such as adhesion reliability, flexibility, and skin comfort. For microneedle formats, the platform will integrate manufacturing-aware design principles to enable reproducible microstructure formation, consistent loading of actives, and reliable dissolution and release under skin-relevant conditions.
- Overall, the proposed technology will establish a unified skin recovery platform capable of generating multiple prototype formats derived from a common scientific and manufacturing foundation. By integrating formulation engineering, patch/microneedle device design, and standardized testing pipelines, the platform will support the development of diversified prototype products with clear performance rationale and practical translational potential, thereby strengthening downstream development readiness and enabling efficient progression toward commercialization-oriented outcomes.

Rationale for Joint R&D Project

- Injuries and inflammation to the skin compromise its structural integrity, affecting millions of people worldwide and creating significant economic and societal burdens for global healthcare systems. In response, natural plant-derived compounds with anti-inflammatory, antioxidant, and antimicrobial properties have been extensively studied in wound care for their ability to support the wound-healing

	process, particularly by reducing prolonged inflammation and promoting tissue regeneration. Growing demand for natural, plant-derived products in healthcare, cosmetics, and advanced wound care is driving industry interest in scientifically validated alternatives to synthetic and steroid-based treatments, due to concerns related to long-term safety, antimicrobial resistance, and regulatory scrutiny. Natural compounds offer a promising pathway, as many plant-based extracts exhibit anti-inflammatory, antioxidant, and antimicrobial properties that align with both clinical requirements and market demand. Academic and research partnerships play a critical role in bridging this capability gap by providing access to fundamental expertise in bioactive compound characterization, mechanism-of-action studies, formulation development, and advanced fabrication technologies such as microneedles, patches, and controlled-release delivery systems. Collaboration with universities and research institutes helps ensure that natural product–based solutions are rigorously validated, reproducible, scalable, and aligned with regulatory and quality standards necessary for commercialization. • This project requires a joint R&D framework because the proposed skin recovery platform spans multiple development steps that must be advanced in parallel, from natural extract preparation and quality control to formulation engineering, patch/microneedle fabrication, and performance validation. It is challenging for an SME to achieve product-ready outcomes within a short timeframe, as progressing from fundamental research to product development exceeds the practical capacity of a single SME. • A major technical challenge is that natural extracts are not “plug-and-play” ingredients. Their performance depends strongly on extraction consistency, marker standardization, stability during storage, and compatibility with polymer matrices. Without a structured QA/QC and standardization workflow, batch-to-batch variability can quickly undermine reproducibility and decision-making during candidate selection and prototype optimization. For this reason, collaboration with a Korean SME partner is essential to support extract preparation (including solid/powder formats where applicable), formulation development, and practical quality management aligned with commercial requirements. • At the same time, patch- and dissolving microneedle-based systems introduce device-level design and testing requirements that go beyond conventional topical formulation work. The platform must be engineered for controlled release and stable on-skin performance, and the microneedle format requires reproducible microfabrication and confirmation of reliable behavior under skin-relevant conditions. These elements must be linked to standardized biological and skin-relevant evaluation methods so that prototype iterations are guided by measurable performance outcomes rather than trial-and-error. • In short, **this international joint R&D framework is designed to ensure that intellectual property is owned by the participating Korean SME**, enabling rapid and direct commercialization alongside technology development. By integrating extraction, formulation, and quality-control expertise with validation and optimization capabilities, the project establishes a realistic development pathway from candidate selection to a diversified prototype portfolio, while maintaining reproducibility, manufacturability, and strong translation potential toward market-ready products.
Key Words	1. Skin recovery patch platform 2. Anti-inflammatory natural actives 3. Dissolving microneedle delivery 4. Barrier restoration and soothing 5. Modular multi-product prototypes

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	• The overarching objective of this R&D project is to develop a scalable skin recovery technology platform that enables reliable incorporation of scientifically supported anti-inflammatory natural extracts into patch- and dissolving microneedle-based delivery formats. Rather than developing a single fixed formulation, the project focuses on establishing transferable formulation, fabrication, and evaluation technologies that can be applied across multiple prototype formats, while maintaining reproducibility, stability, and product feasibility toward market-ready outcomes. • To achieve this objective, the project will first establish a candidate evaluation framework for anti-inflammatory natural extracts suitable for skin recovery applications. A set of candidate extracts will be screened using standardized biological efficacy and safety criteria, together with early-stage formulation feasibility indicators such as polymer compatibility and stability behavior. Candidates will be selected from commonly used anti-inflammatory botanical sources (e.g., Hydrangea serrata, Centella asiatica, Glycyrrhiza spp., and related botanicals), and the final lead candidate will be determined based on reproducible performance and product-development feasibility. • A second objective is to develop robust formulation and stability strategies that address common translational barriers of natural extracts, including batch-to-batch variability, storage-related degradation, and undesired phase behavior in polymer matrices. The project will develop practical approaches for extract standardization and QA/QC, supported by marker-based characterization where appropriate, to ensure that selected candidates can be consistently formulated and reproduced for prototype development. Formulation design will prioritize solid extract–polymer systems when applicable, including protective or passivation strategies to preserve marker integrity and reduce crystallization or instability during processing and storage. • A third objective is to engineer and fabricate complementary product formats, including film- or hydrogel-based adhesive patches and dissolving microneedle patches, as a unified delivery platform. Patch prototypes will be designed to support practical on-skin use, emphasizing stable adhesion, wearability, skin comfort, and controlled local exposure of natural actives. In parallel, dissolving microneedle formats will be developed as an advanced product line to enhance delivery efficiency by enabling controlled interaction beyond the stratum corneum, while maintaining minimally invasive and user-friendly application. • Finally, the project aims to establish standardized testing and validation pipelines that link formulation/device parameters with measurable performance outcomes. These evaluation workflows will include controlled release characterization, stability screening under relevant storage and packaging conditions, and skin-relevant performance assessments to guide iterative optimization. The final deliverable of this R&D project will be a diversified prototype portfolio derived from a common scientific and manufacturing foundation, including multiple patch and microneedle-based prototypes supported by reproducible evidence packages and scale-up oriented development considerations.			
R&D Target Specifications	• The target performance specifications of this R&D project are defined to ensure that the proposed patch- and microneedle-based skin recovery platform delivers measurable improvements in soothing performance, recovery-support functionality, delivery consistency, and product stability across multiple prototype formats. These specifications will serve as practical decision criteria for candidate down-selection and iterative prototype optimization, with emphasis on reproducibility and manufacturability. • Biological performance targets will be applied for candidate screening and lead selection. Lead candidates among the natural extracts will be required to demonstrate clear inflammation-modulating effects in standardized screening models, targeting at least ≥40% reduction of key inflammatory markers (e.g., TNF-α and IL-6) and ≥40% reduction of nitric oxide (NO) production compared with inflammation-induced controls (6). This threshold will serve as a practical stage-gate decision			

	criterion to prioritize candidates with robust and reproducible effect sizes suitable for downstream formulation and prototype development. To further strengthen lead prioritization, candidates achieving ≥50% reduction of key inflammatory markers under reproducible conditions will be considered as top-tier leads, where applicable (6). Biocompatibility will be ensured by maintaining ≥85% cell viability at the intended active concentration range, supporting reliable incorporation into skin-contact product formats. • To strengthen evidence of skin recovery benefits beyond cell-only endpoints, additional targets will be applied using skin-relevant evaluation systems where appropriate. These include achieving a statistically significant reduction of irritation-related inflammatory biomarkers (e.g., IL-1α), and demonstrating recovery-support trends such as ≥15% improvement in hydration-related indicators compared with blank patch controls (7), depending on the evaluation model and measurement method used. In addition, the project will aim to demonstrate accelerated barrier recovery kinetics compared with blank patch controls, where measurable within the selected evaluation systems. • From a formulation and product performance perspective, patch prototypes will be designed to meet practical use requirements for wearability and user compliance. Adhesion and wearability will be engineered to enable stable attachment under normal daily-use conditions, with minimal edge lifting or loss of contact. Controlled-release behavior will be optimized to maintain sustained local exposure of natural actives throughout the intended wear period, while minimizing burst-dominant release profiles that can shorten the effective duration of exposure. Release kinetics will be tuned to support a practical onset of activity followed by continued delivery, depending on prototype format and intended use conditions. In parallel, formulations will prioritize long-term stability of natural actives through appropriate solid extract–polymer matrix designs, including protective or passivation strategies where necessary, to preserve marker integrity and prevent undesirable crystallization, phase separation, or other instability during processing, storage, and use. • For microneedle-based prototypes, performance specifications will focus on reproducible geometry, mechanical reliability, and delivery enhancement. Target microneedle lengths will be optimized within a sub-millimeter design space (e.g., 300–800 μm), recognizing that insertion performance is skin-model dependent and that shorter needles may exhibit limited penetration in certain test systems (8). Final geometries will be determined based on insertion validation and delivery assessments in skin-relevant models. Mechanical integrity will be evaluated through compression testing to confirm structural robustness during handling and application, and insertion performance will be verified to ensure reliable penetration and reproducible delivery under skin-relevant conditions. Dissolving behavior will be optimized for practical use to support consistent dissolution and active release, and microneedle-enabled delivery will be compared against non-microneedle patch formats to confirm clear improvement in marker delivery. • Finally, stability and manufacturing readiness specifications will be established to support downstream product development. Extract formulations and final prototypes will be required to demonstrate acceptable storage stability under relevant packaged conditions, supported by marker-based monitoring where appropriate. To address inherent variability of natural extracts, the project will implement QA/QC criteria for extract preparation and batch consistency, and will define prototype reproducibility requirements for critical physical parameters (e.g., patch dimensions, weight uniformity, thickness consistency, and microneedle geometry consistency). Collectively, these specifications will ensure that the developed platform is not only scientifically supported but also practically scalable toward market-ready prototype outcomes.
R&D Plan	The proposed R&D project will be carried out over a three-year period using a stage-gated development strategy designed to systematically advance from candidate natural extract evaluation to formulation optimization and multi-format prototype development. The plan integrates formulation engineering, patch and microneedle fabrication, and biological performance validation, enabling evidence-based down-selection of lead candidates and progressive de-risking toward market-ready prototype outcomes. **Year 1: Candidate Evaluation and Platform Feasibility Establishment** • Year 1 will establish the scientific and technical foundation of the skin recovery platform. Multiple

botanical extracts with reported anti-inflammatory potential (e.g., Hydrangea serrata and related natural extracts) will be considered. With support from an industrial partner experienced in natural extract processing and formulation, baseline extract quality parameters will be defined, including marker compound characterization, batch consistency assessment, and stability considerations relevant to formulation development. In addition, early formulation feasibility checks will be conducted (e.g., solubility/compatibility in representative polymer matrices and preliminary stability screening).

- Concurrently, standardized in vitro assays will be conducted to evaluate inflammation-modulating efficacy and cytocompatibility in skin-relevant cell models. Screening will include quantification of cytokine/inflammatory marker modulation under induced inflammatory conditions, together with basic irritation-related biomarkers where applicable. These datasets will support the first stage-gate decision to down-select one or two lead candidates based on reproducible biological performance and formulation feasibility.
- Alongside candidate screening, film-based adhesive patch prototypes and dissolving microneedle patch architectures will be designed and fabricated to assess compatibility with solid natural extract formulations. Microfabrication approaches such as micromilling or laser-assisted patterning will be used to generate master molds, followed by formation of polymer-based passivation/protective layers using biocompatible materials. Solid extract-polymer formulations will be incorporated using embedded or filling approaches to evaluate manufacturability, structural integrity, and initial release behavior. Feasibility will be demonstrated through in vitro proof-of-concept studies, including controlled release testing under skin-relevant conditions and initial stability observations after fabrication.

Year 2: Formulation Optimization, Multi-Format Prototype Development, and In vivo Performance Tests

- Year 2 will focus on optimization of formulation and device performance for selected lead candidate, with development of multiple product formats in coordinated work streams. For patch systems, hydrogel and film prototypes will be refined by optimizing polymer matrices, extract loading strategies, adhesion behavior, controlled-release characteristics, and stability under relevant storage/packaging conditions. Systematic optimization will be supported by iterative formulation trials and release profiling to tune early availability and sustained exposure during intended use.
- In tandem, dissolving microneedle systems will be further developed to enhance interaction beyond the stratum corneum, optimizing geometry, material composition, and fabrication parameters to ensure reproducibility, mechanical robustness, and consistent dissolution behavior. Microneedle insertion feasibility and functional behavior will be verified using standardized skin-relevant models, and delivery performance will be compared against non-microneedle patch formats.
- At the same time, in vivo animal study will be performed to validate through iterative testing cycles, linking performance outcomes to formulation/device parameters. A second stage-gate decision at the end of Year 2 will confirm final candidate and prototype formats to be advanced.

Year 3: Validation, Reproducibility, and Scale-Up Readiness

- Year 3 will consolidate performance evidence, finalize a diversified prototype portfolio, and address scale-up readiness. Selected prototypes will undergo final optimization to improve reproducibility, usability, and stability. Where relevant, hybrid formats combining microneedles with backing patches will be explored to integrate enhanced delivery with on-skin functionality and user comfort.
- Meanwhile, expanded stability evaluations will be performed under packaged storage conditions with marker-based monitoring, supported by physical stability checks (appearance, phase behavior, and consistency of release profiles after storage). Manufacturing-oriented assessments will establish reproducible fabrication workflows, define acceptable process tolerances, and identify critical quality attributes for both patch and microneedle formats. The industrial partner will support process transfer considerations including batch consistency, raw-material handling, and practical manufacturing constraints.
- By the end of Year 3, the project will deliver a validated prototype portfolio derived from a unified platform, potentially including film patches, hydrogel patches, dissolving microneedle patches, and hybrid formats, supported by reproducible performance datasets and defined formulation/fabrication specifications.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• The research institute will act as the scientific and technical lead of the joint R&D project, ensuring that candidate selection and prototype development are driven by reproducible evidence rather than trial-and-error. The institute will be responsible for building the core evaluation framework of the skin recovery platform and generating validated datasets that can support clear decision-making throughout the stage-gated development process. • A key responsibility will be the biological screening and down-selection of candidate anti-inflammatory natural extracts for skin recovery use. The institute will design and run standardized in vitro assays under inflammation-induced conditions, quantify relevant endpoints (e.g., cytokines and inflammation-related markers), and conduct compatibility screening using skin-relevant test systems. These results will be used to rank candidates and recommend lead extract(s) for advancement into prototype development. • The institute will also lead the technical development of the delivery platforms, including formulation engineering and prototype fabrication. This will cover the design of solid extract-polymer systems, optimization of film and hydrogel patch formats, and development of dissolving microneedle prototypes using available microfabrication capability. The institute will establish practical fabrication workflows, evaluate structural integrity and reproducibility of prototypes, and assess functional behaviors such as controlled release, microneedle dissolution, and insertion feasibility in skin-relevant evaluation models. • To connect product engineering with real performance, the institute will operate standardized validation pipelines that link formulation/device parameters to biological outcomes. Through iterative test cycles, the institute will support prototype optimization, compare performance across patch and microneedle formats, and generate a structured evidence package that supports progression toward product-ready prototypes. • To support downstream scale-up and transfer, the institute will provide technical documentation and validated performance datasets. Key outputs will include candidate screening results, prototype fabrication and characterization records, and evaluation reports that enable efficient coordination with the industrial partner toward manufacturing-oriented development and commercialization planning.
Expected R&R of Korean SME Partner	• The Korean SME partner will serve as the industrial development and translation lead of the joint R&D project, with primary responsibility for natural extract preparation, formulation practicality, and product-oriented feasibility toward scalable prototype outcomes. The partner's role is essential to ensure that the proposed skin recovery platform is not only scientifically validated, but also manufacturable and consistent under real product-development constraints. • The SME partner will support candidate extract readiness by enabling reproducible extraction and processing conditions, including practical input on preparation routes, preferred dosage forms, and marker-based quality control strategies to address botanical variability. Where applicable, the partner will contribute to batch consistency evaluation and provide traceable extract samples for screening and prototype fabrication. • The SME partner will conduct animal studies to validate the wound-healing and anti-inflammatory performance of the patch system developed by the Canadian research team. Based on in vivo data, the SME will optimize formulations and take responsibility for prototype fabrication. Leveraging industrial know-how, the partner will support selection of practical polymer systems, additives, and processing conditions compatible with natural extracts, while mitigating instability risks such as crystallization, phase separation, or storage-related degradation. The partner will also provide input on process robustness, material handling, packaging, and key QA/QC checkpoints relevant to scale-up and commercialization. • Overall, the Korean SME partner will ensure the joint R&D program delivers development-ready prototypes by supporting extract standardization, manufacturing feasibility, quality consistency, and practical product translation across the three-year timeline.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• This project is expected to deliver a validated and scalable skin recovery technology platform that enables systematic development of patch- and microneedle-based products incorporating anti-inflammatory natural actives, with a strong emphasis on reproducible performance and scientifically interpretable mechanisms. A central outcome will be the establishment of an evidence-driven R&D framework that bridges biological effect, material formulation, and device design within a single development pipeline. Rather than treating natural extract performance as a formulation-only problem, the proposed program will directly connect extract quality parameters, polymer matrix behavior, release kinetics, and skin-delivery architecture to measurable anti-inflammatory and recovery-related endpoints. This integrated approach represents a key scientific advantage because it enables rational optimization based on mechanistic and quantitative datasets, reducing the common trial-and-error cycles that often limit the translation of botanical actives into reliable skin-contact products. • The project will generate a portfolio of prototype formats derived from a unified scientific and manufacturing foundation, including film-based recovery patches, hydrogel-based soothing patches, dissolving microneedle systems, and hybrid formats were beneficial. Importantly, prototypes will not be treated as isolated outputs, but as experimental platforms used to build structure–function relationships across three domains: (i) biological performance (inflammation modulation and irritation-associated markers under controlled screening models), (ii) formulation performance (stability and compatibility of natural actives within solid extract–polymer matrices, including mitigation of undesired crystallization or phase instability), and (iii) delivery performance (controlled local exposure through patch formats and enhanced delivery efficiency through microneedle-enabled architectures). Each prototype will be supported by reproducible datasets that enable objective comparison across formats and help identify the most robust design choices for downstream translation. • A major scientific contribution of this project will be the generation of standardized evaluation pipelines and decision criteria that enable rigorous candidate down-selection from multiple natural extracts. Natural extracts are intrinsically complex and variable, and their development risk is often underestimated during early R&D stages. The platform will address this constraint by establishing marker-based characterization approaches, extract handling guidance, batch consistency checkpoints, and standardized biological screening conditions that allow fair benchmarking across candidates. These outcomes will strengthen not only the internal progression of the project but also future applicability of the platform, by allowing additional candidate extracts to be evaluated and integrated without rebuilding the full evaluation system. The program will also deliver practical formulation engineering solutions for stabilizing natural extracts in polymer matrices and protecting sensitive actives during processing and storage through appropriate matrix design and passivation strategies, thereby increasing the feasibility of real product development using botanical materials. • The impact of the proposed joint R&D program lies in resolving a well-known translational gap between natural-active discovery and product-ready skin delivery technologies. Skin irritation and recovery needs extend beyond cosmetic discomfort and occur widely due to friction, dryness, minor abrasion, thermal stress, environmental exposure, and procedure-related irritation. Yet many conventional topical products remain limited by short residence time, inconsistent dosing, low delivery efficiency, and high dependence on user application behavior, which together produce variable outcomes in real-world use. By advancing wearable delivery formats that can provide controlled and sustained local exposure of natural actives, the proposed platform increases the feasibility of fast-acting and user-friendly recovery solutions supported by measurable biological evidence. The microneedle component further strengthens translational potential by expanding the feasible delivery space beyond conventional topical approaches, while maintaining compatibility with practical and minimally invasive skin-contact designs. • From a commercialization perspective, <u>this project is structured to achieve rapid market translation by ensuring that the resulting intellectual property is owned by the Korean SME partner.</u> This IP ownership model enables technology development and commercialization to proceed in parallel, rather than sequentially. By combining the Canadian research institute's expertise in drug delivery

	systems and formulation development with the Korean SME's specialized capabilities in raw material production and processing, in vivo performance validation through animal studies, and product manufacturing technologies, the project establishes a strong foundation for developing novel natural-active-based therapeutic products for wound healing and skin recovery. • The platform-oriented development approach enables modular formulation and device integration rather than one-off product design. An integrated framework encompassing extract standardization, formulation engineering, device fabrication, biological performance validation, and manufacturability assessment supports faster iteration, improved reproducibility, and diversified product positioning. Within this joint structure, the Canadian research institute will lead biological validation, standardized performance testing, and engineering evaluation of the drug delivery platform, while the Korean SME will drive formulation practicality, QA/QC considerations, animal-based efficacy validation, and prototype manufacturing. • This collaborative structure supports the generation of product-ready evidence packages and process-ready know-how under an IP regime optimized for commercialization, thereby strengthening technology readiness, reducing scale-up risk, and accelerating downstream translation toward market-ready natural-product-based wound healing therapeutics.

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